ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ELECTRICAL MACHINES

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 2nd year Electrical Machines paper 1 · practice a.

Written by Academic Central for practice, following electrical machines modules at South African universities (e.g. University of Pretoria ELX 311 Electrical Machines and EIR 211 Electrical Engineering: magnetic circuits, transformers, per-unit system, three-phase circuits, DC and induction motors). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method and answers that differ only through rounding. Deduct one mark only once per question for missing units.

QUESTION 1

1.1 MMF: the "driving force" of a magnetic circuit, $F = NI$, unit ampere-turn (A·t) ✓ (4)
Flux Φ: the total magnetic field through an area, unit weber (Wb) ✓
Flux density $B = \Phi/A$, unit tesla (T) ✓
Reluctance $S = F/\Phi = l/(\mu_0\mu_r A)$, unit A·t/Wb ✓

1.2.1 $S = l / (\mu_0\mu_r A) = 0.4 / (4\pi \times 10^{-7} \times 1\,000 \times 5 \times 10^{-4})$ ✓ = 636 620 A·t/Wb ✓ (4)
$F = \Phi S = 0.5 \times 10^{-3} \times 636\,620 = 318.31$ A·t ✓
$I = 318.31 / 200 = 1.59$ A ✓

1.2.2 $B = \Phi/A = 0.5 \times 10^{-3} / 5 \times 10^{-4} = 1$ T ✓ (1)

1.3 $S_{gap} = 10^{-3} / (4\pi \times 10^{-7} \times 5 \times 10^{-4})$ ✓ = 1 591 549 A·t/Wb ✓ (4)
$S_{total} = 636\,620 + 1\,591\,549 = 2\,228\,169$ A·t/Wb ✓
$I = 0.5 \times 10^{-3} \times 2\,228\,169 / 200 = 5.57$ A ✓

1.4 Hysteresis loss: energy used to keep reversing the magnetism of the iron each cycle ✓; reduced by using soft magnetic material such as silicon steel, with a narrow hysteresis loop ✓ (4)
Eddy-current loss: currents induced in the core itself heat it ✓; reduced by laminating the core (thin, insulated sheets) and by using high-resistivity steel ✓

1.5 $e = N\,d\Phi/dt = 500 \times 2 \times 10^{-3} / 0.01$ ✓✓ = 100 V ✓ (3)

[20]

QUESTION 2

2.1 AC in the primary sets up an alternating flux in the iron core ✓. The flux links the secondary winding ✓ and induces an EMF in it (Faraday's law), with $V_1/V_2 = N_1/N_2$ ✓ (3)

2.2.1 $N_1/N_2 = 230/12 = 19.17$ ✓ (3)
$N_2 = 1\,200 \times 12 / 230$ ✓ = 62.6, so 63 turns ✓

2.2.2 $I_1 = 50 / 230 = 0.22$ A ✓ $I_2 = 50 / 12 = 4.17$ A ✓ (2)

2.3 $E = 4.44\,f\,N\,\Phi m$ ✓ (3)
$\Phi m = 230 / (4.44 \times 50 \times 1\,200)$ ✓ = 0.86 mWb ✓

2.4 Correct circuit: series R and X (winding resistance and leakage reactance), shunt R_c and X_m at the input, ideal transformer ✓✓ (4)
R: copper losses in the windings; X: leakage flux ✓
R_c: core (iron) losses; X_m: magnetising current ✓

2.5.1 Output = 10 000 × 0.8 = 8 000 W ✓ (3)
Losses = 200 + 300 = 500 W ✓
$\eta = 8\,000 / 8\,500 = 94.12\%$ ✓

2.5.2 Output = 4 000 W; copper loss = $(0.5)^2 \times 300 = 75$ W ✓ (3)
Losses = 200 + 75 = 275 W ✓
$\eta = 4\,000 / 4\,275 = 93.57\%$ ✓

2.5.3 Maximum when copper loss = core loss: $x^2 \times 300 = 200$ ✓ (2)
$x = \sqrt{(200/300)} = 0.82$ of full load ✓

2.6 Regulation = $(V_{NL} - V_{FL}) / V_{FL} = (200 - 192)/192$ ✓ = 4.17% ✓ (accept 4% if divided by V_{NL}) (2)

[25]

QUESTION 3

3.1 Star: $V_{ph} = 400/\sqrt{3} = 230.94$ V ✓; $I_L = I_{ph}$ ✓ (4)
Delta: $V_L = V_{ph}$ ✓; $I_L = \sqrt{3}\, I_{ph}$ ✓

3.2 $I_L = P / (\sqrt{3}\, V_L \cos\varphi) = 15\,000 / (\sqrt{3} \times 400 \times 0.85)$ ✓ = 25.47 A ✓ (5)
$S = 15 / 0.85 = 17.65$ kVA ✓
$Q = S \sin\varphi = 17.65 \times 0.527$ ✓ = 9.30 kVAr ✓

3.3 $I_{base} = 10\,000 / (\sqrt{3} \times 400) = 14.43$ A ✓ (4)
$Z_{base} = 400^2 / 10\,000 = 16\ \Omega$ ✓
$Z_{pu} = 0.8 / 16$ ✓ = 0.05 pu ✓

3.4 Any TWO: per-unit values of similar machines fall in a narrow range, so errors are easy to spot; transformer impedances are the same on both sides, so turns ratios disappear from the calculations; the numbers are easy to compare between machines of different sizes ✓✓ (2)

[15]

QUESTION 4

4.1 The commutator and brushes connect the external DC supply to the rotating armature coils ✓, and reverse the current in each coil as it passes the neutral position ✓, so the torque on the armature always acts in the same direction ✓ (3)

4.2.1 $E_b = V - I_a R_a = 220 - 20 \times 0.5$ ✓ = 210 V ✓ (2)

4.2.2 $P = E_b I_a = 210 \times 20 = 4\,200$ W ✓ (3)
$\omega = 1\,000 \times 2\pi/60 = 104.72$ rad/s; $T = 4\,200 / 104.72$ ✓ = 40.11 N·m ✓

4.2.3 $E_b = 220 - 30 \times 0.5 = 205$ V ✓ (3)
$n \propto E_b$ (same flux): $n = 1\,000 \times 205/210$ ✓ = 976.19 r/min ✓

4.3 $I_{start} = 220 / 0.5 = 440$ A ✓ (4)
There is no back EMF at standstill, so the very large current can damage the windings and commutator and trip the supply ✓
$R_{total} = 220 / 40 = 5.5\ \Omega$ ✓, so $R_{start} = 5.5 - 0.5 = 5\ \Omega$ ✓

4.4 Armature voltage control: changing the voltage changes the speed below base speed ✓ (3)
Field weakening: reducing the field current raises the speed above base speed ✓
A robot motor driver normally uses armature voltage control by pulse-width modulation (PWM) through an H-bridge ✓

4.5 Series motor: very high starting torque, speed falls a lot as load rises (must not run on no load), e.g. traction, cranes, starter motors ✓
Shunt motor: almost constant speed as load changes, e.g. conveyors, machine tools, fans ✓ (2)

[20]

QUESTION 5

5.1 Three-phase currents in the stator windings produce a magnetic field that rotates at synchronous speed ✓. The field cuts the rotor bars and induces currents in them ✓. These currents in the magnetic field produce a force, so the rotor turns, always slightly slower than the field (slip) ✓ (3)

5.2 $n_s = 120f/p = 120 \times 50 / 4 = 1\,500$ r/min ✓
$s = (1\,500 - 1\,440)/1\,500$ ✓ $= 0.04$ (4%) ✓
$f_r = sf = 0.04 \times 50 = 2$ Hz ✓ (4)

5.3 $\omega = 1\,440 \times 2\pi/60 = 150.80$ rad/s; $T = 7\,500 / 150.80$ ✓ $= 49.74$ N·m ✓
$P_{in} = \sqrt{3} \times 400 \times 15 \times 0.85$ ✓ $= 8\,833.46$ W ✓
$\eta = 7\,500 / 8\,833.46 = 84.90\%$ ✓ (5)

5.4 Rotor copper loss $= s \times P_{ag} = 0.04 \times 8\,000$ ✓ $= 320$ W ✓
$P_{mech} = 8\,000 - 320 = 7\,680$ W ✓ (3)

5.5 DOL: full voltage at once, simple and cheap, but the starting current is 6 to 8 times full-load current ✓
Star-delta: starts in star with $1/\sqrt{3}$ of the voltage on each winding, so the starting current and torque fall to one third ✓; then it switches to delta; suitable only for light starting loads ✓ (3)

5.6 $n_s = 120f/p$, so $f = 1\,200 \times 4 / 120 = 40$ Hz ✓
$V = 400 \times 40/50 = 320$ V ✓ (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE